Computational PDEs: Classical Approaches and Physics-Informed Learning

Day 3 — Doubly Connected Domains

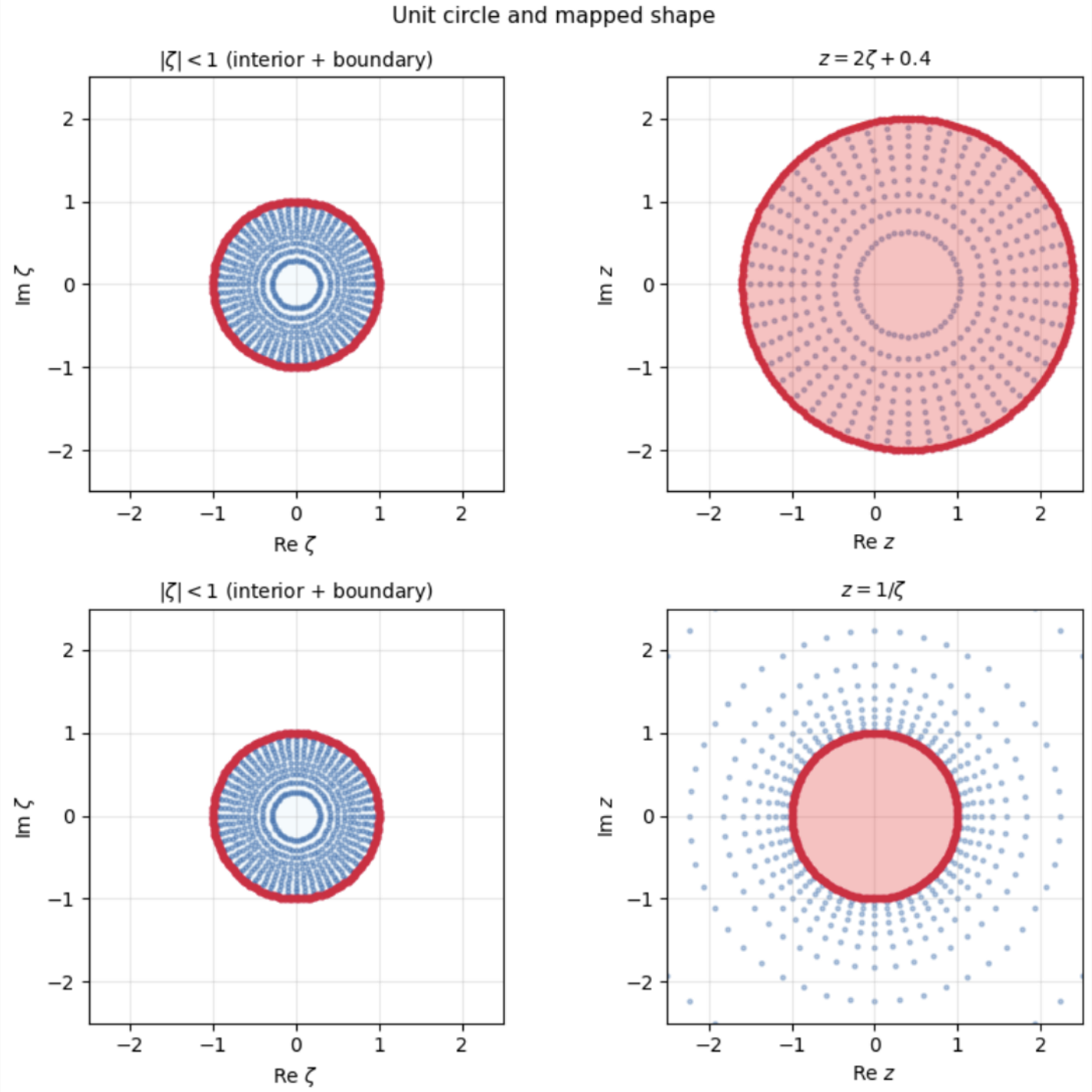
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Exercise (simple conformal mapping)

- $$z = g(\zeta) = 2\zeta + 0.4 \tag{1}$$
- $$z = g(\zeta) = 1/\zeta \tag{2}$$


Unit circle and mapped shape
$|\zeta| < 1$ (interior + boundary)
$z = 2\zeta + 0.4$
$|\zeta| < 1$ (interior + boundary)
$z = 1/\zeta$
Im ζ
Re ζ
Im z
Re z

Exercises: what is the conformal map for the following domains?

- $$z = g(\zeta) = \frac{1}{2}\left(\zeta + \frac{1}{\zeta}\right) \tag{3}$$
- $$z = g(\zeta) = \frac{\zeta - a}{\zeta - b}\frac{\zeta - 1/\overline{a}}{\zeta - 1/\overline{b}} \tag{4}$$
- $$z = g(\zeta) = \frac{|b|}{|a|}\frac{\zeta - a}{\zeta - b}\frac{\zeta - 1/\overline{b}}{\zeta - 1/\overline{a}} \tag{5}$$

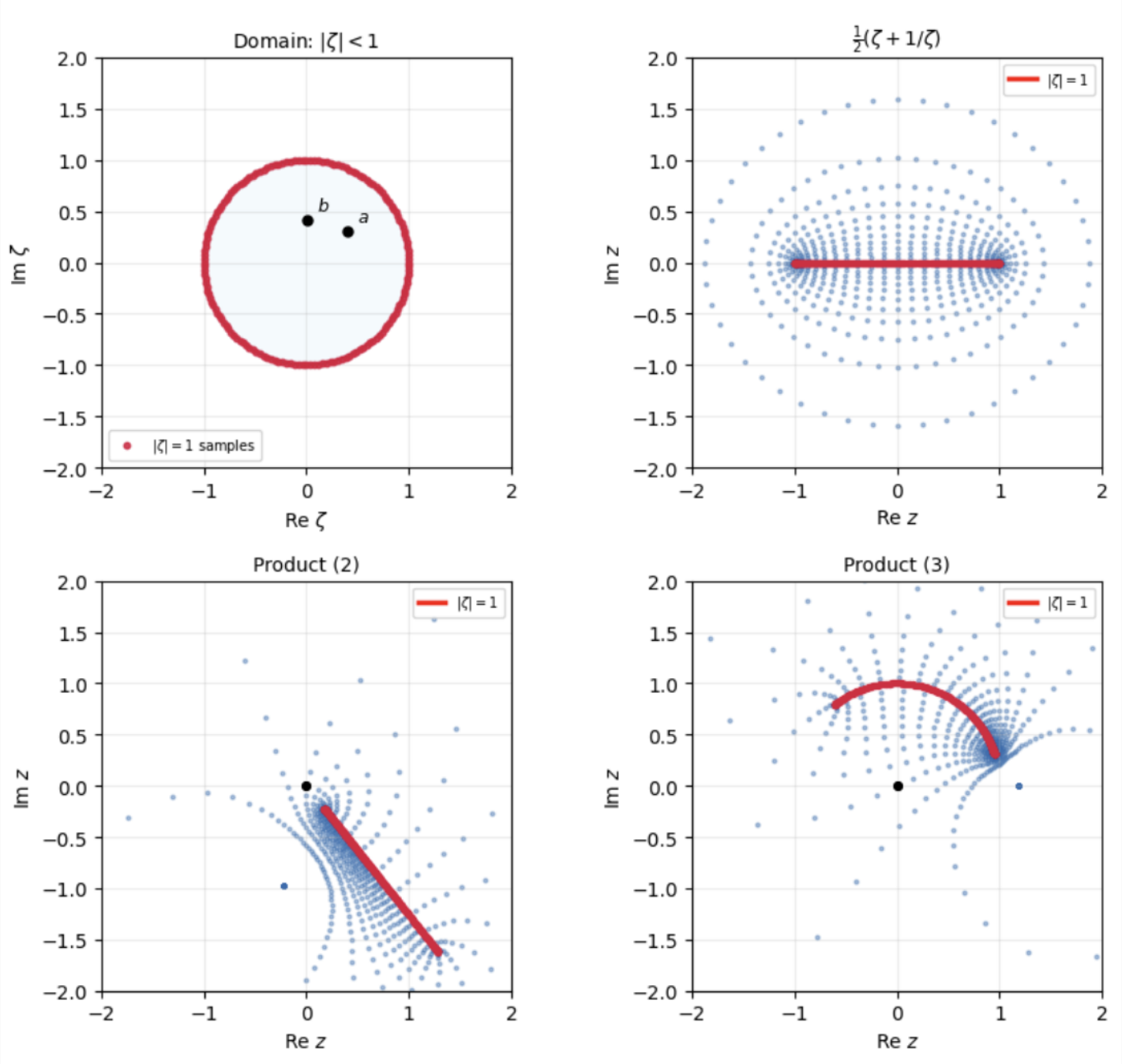
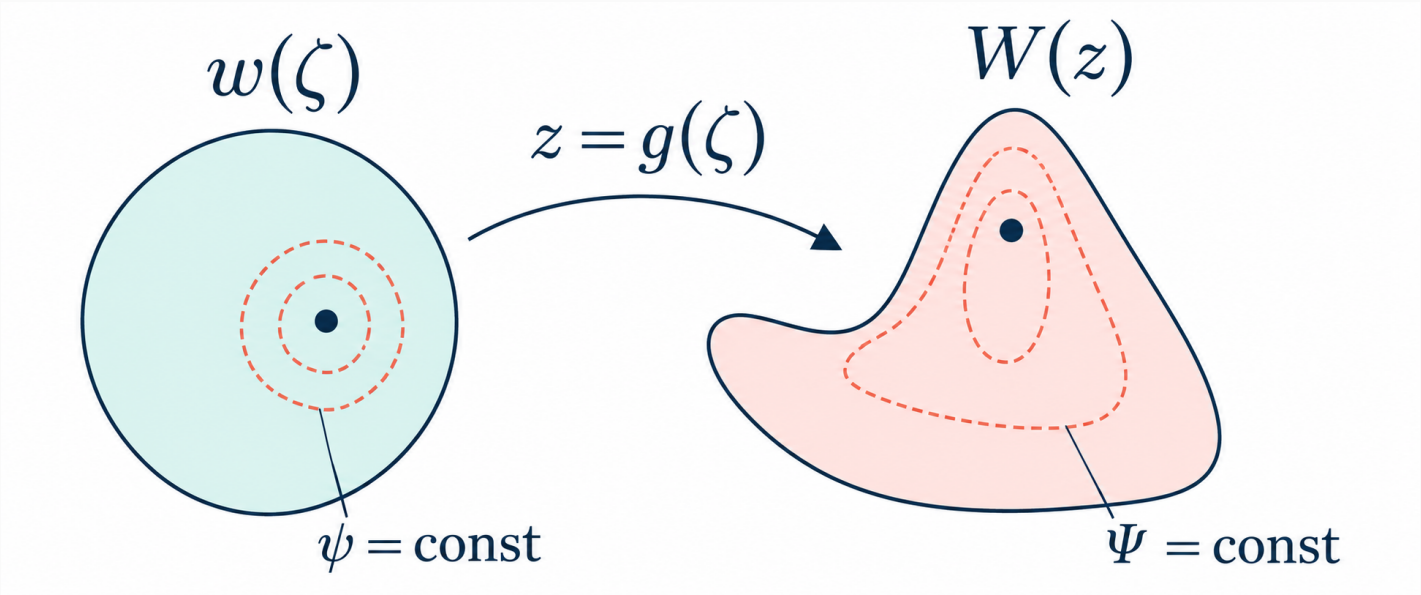


Figure 1: 03_day3_conformal_map_exercises.ipynb

From Tuesday

Tuesday placed one reflected point. A hole is not removed by a map to the disk.

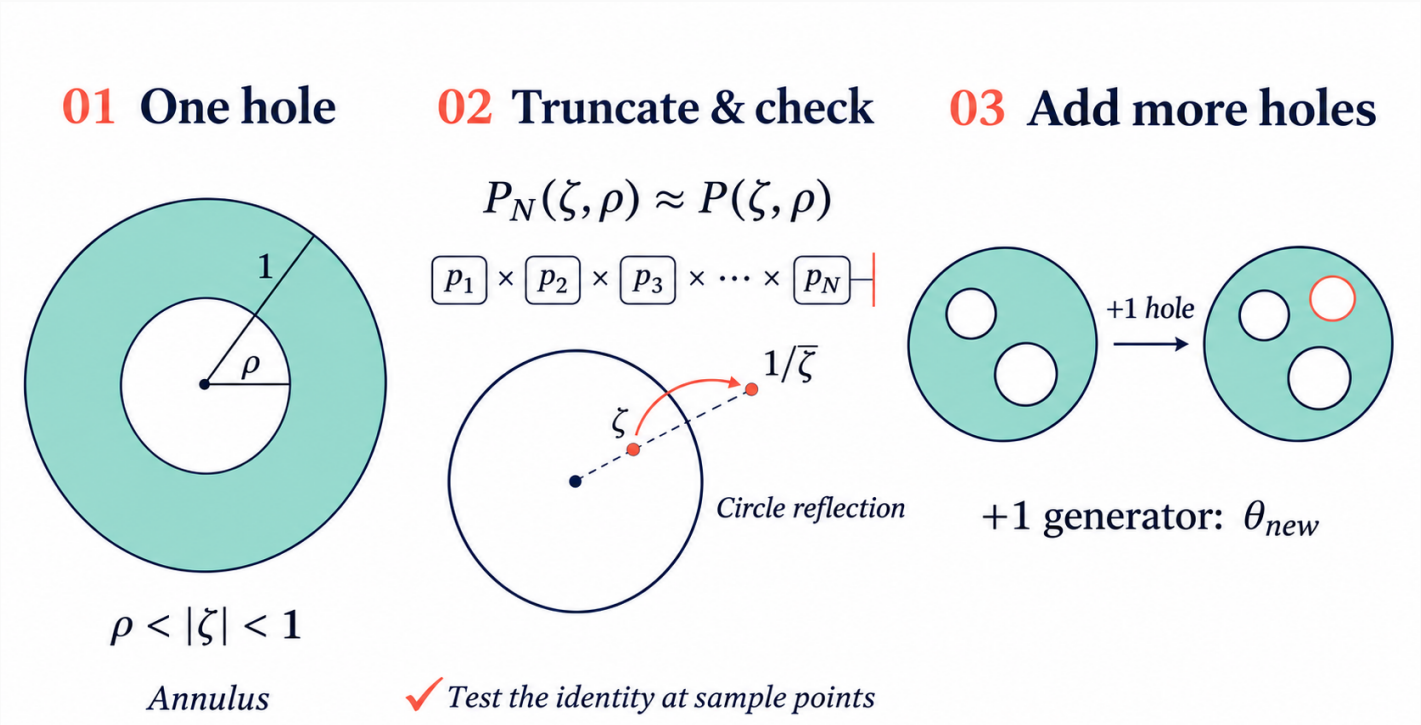


We are going to extend this theory to multiply connected domains.

Aims

After today you should be able to

1. Annulus, with inner radius ρ, as the domain with one hole;
2. truncate a special function $P(\zeta, \rho)$ and evaluate the inversion identity at sample points;
3. say that further holes append one generator (Advanced topic)



A $\log$ singularity inside the unit disk

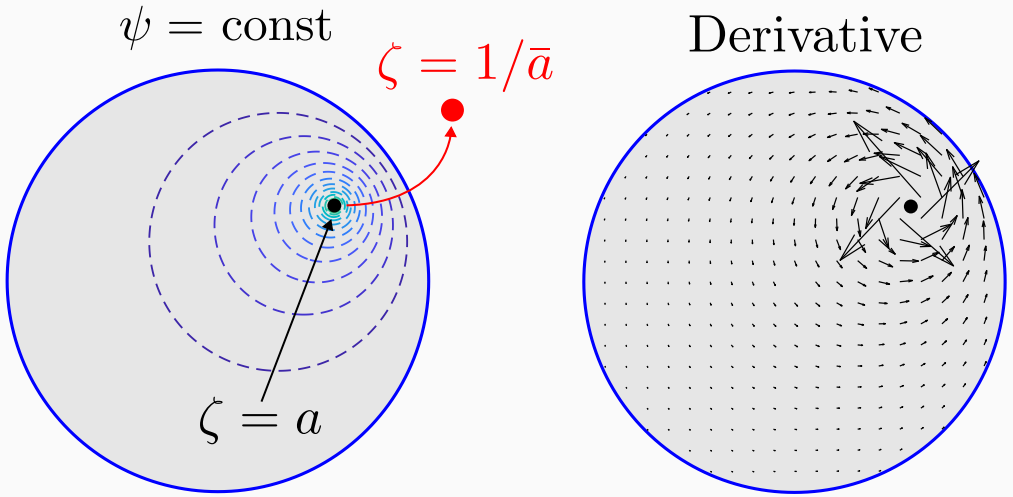
Suppose a $\log$-type singularity sits at $\zeta = a$ inside the unit disk. Impose

$$\psi = 0 \text{ on } \partial\mathbb{D} \tag{6}$$

($\mathbb{D}$: unit disk; one chosen level of ψ on the boundary). Then

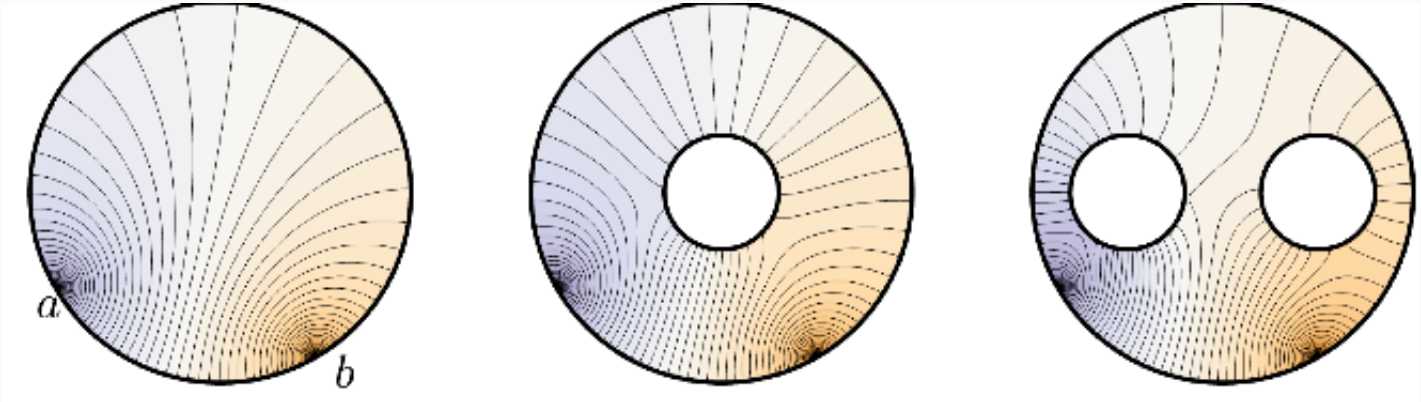
$$w(\zeta) = \phi + \mathrm{i}\psi = \frac{1}{2\pi\mathrm{i}} \log\left(\frac{\zeta - a}{|a|(\zeta - 1/\overline{a})}\right)$$

The **image point in the unit circle** is $\zeta = 1/\overline{a}$.



Riemann mapping theorem (multiply connected domains)

- The **Riemann mapping theorem** extends to **multiply connected domains**.
- There is a conformal map between a simply connected domain and the unit disk.
- A doubly connected domain corresponds to an **annulus** (a domain with one hole).
- A triply connected domain corresponds to a **unit disk with two holes**.



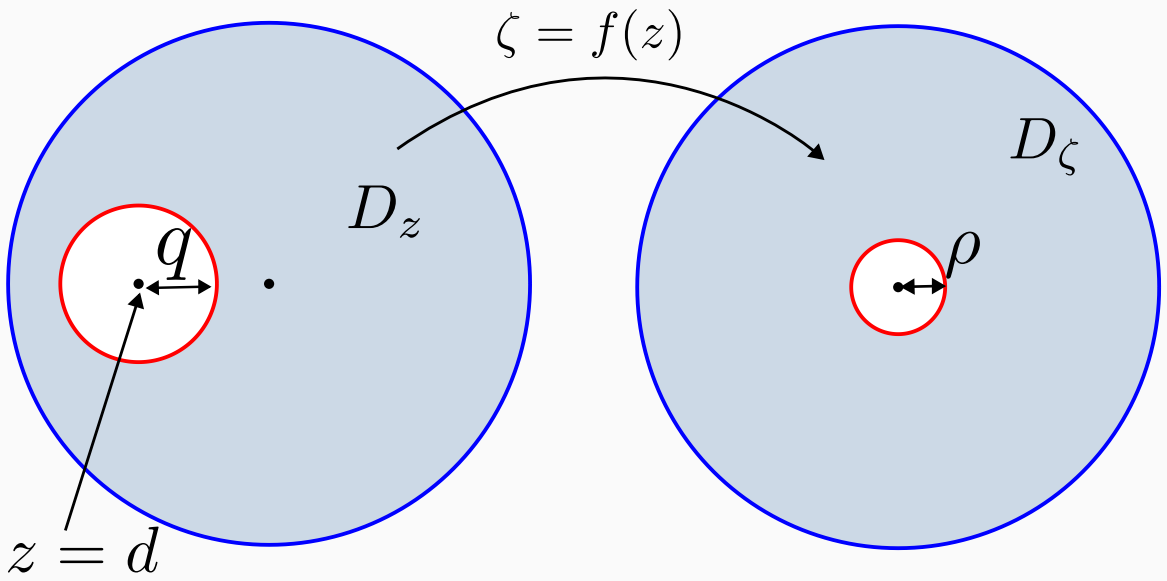
Goal from here

Build the theory on the domain above (a circular canonical domain).

From simply connected to doubly connected domains

Doubly connected domains

- For simply connected domains the unit disk was the starting point.
- For doubly connected domains it is the **annulus**.
- By an **automorphism**, the centre of the inner circle can be mapped to the origin. (Next page)
- The parameter of the annulus is the inner radius ρ (the modulus).
- Domain $D_\zeta = \{\zeta \,|\rho < |\zeta| < 1\}, \quad 0 < \rho < 1.$



Automorphisms

$$\zeta = e^{\mathrm{i}\alpha} \frac{z - z_0}{\overline{z_0} z - 1}, \quad \alpha \in \mathbb{R},\ d \in (D_z \backslash \partial D_z) \tag{7}$$

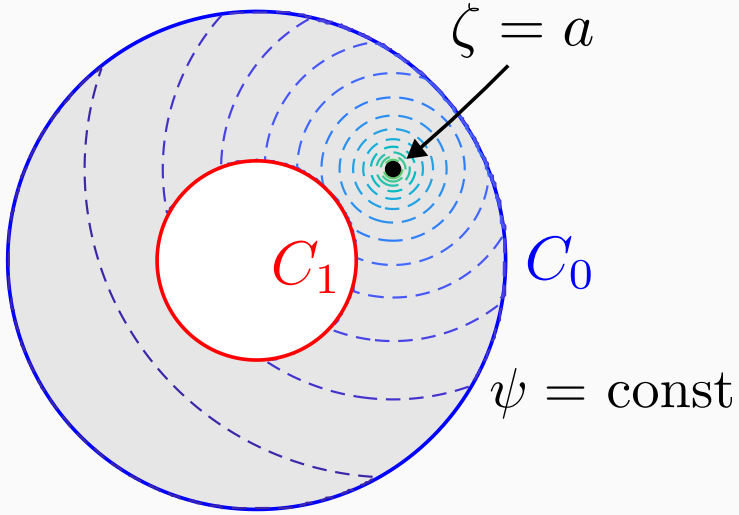
where

$$z_0 = \frac{1 + c^2 - r^2 - \sqrt{(1 + c^2 - r^2)^2 - 4c^2}}{2c} \tag{8}$$

Then a circle inside the unit disk, with center $z = c > 0$ and radius r, is mapped to a circle centered at the origin.

Construction of complex potential

- Suppose a point vortex is located at $\zeta = a$ in the annulus.
- The unit circle C_0 and the inner circle C_1 are to be streamlines.
  $\rightarrow$ **There is no flux into the domain.**
- We want ψ to be constant on both C_0 and C_1.
  $\rightarrow$ Apply the idea of the method of images infinitely often.

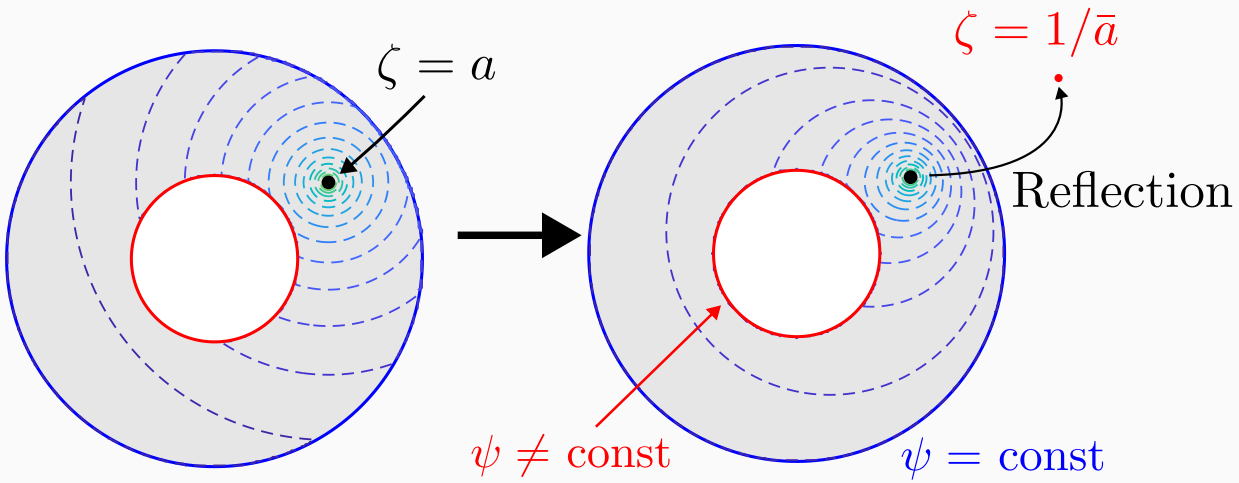


Construction of the complex potential

- For $\zeta = a$, consider the potential that includes the image $1/\overline{a}$ in the unit circle:

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}} (\log(\zeta - a) - \log(\zeta - 1/\overline{a})) \tag{9}$$

- The boundary condition is satisfied on C_0, but not on C_1.

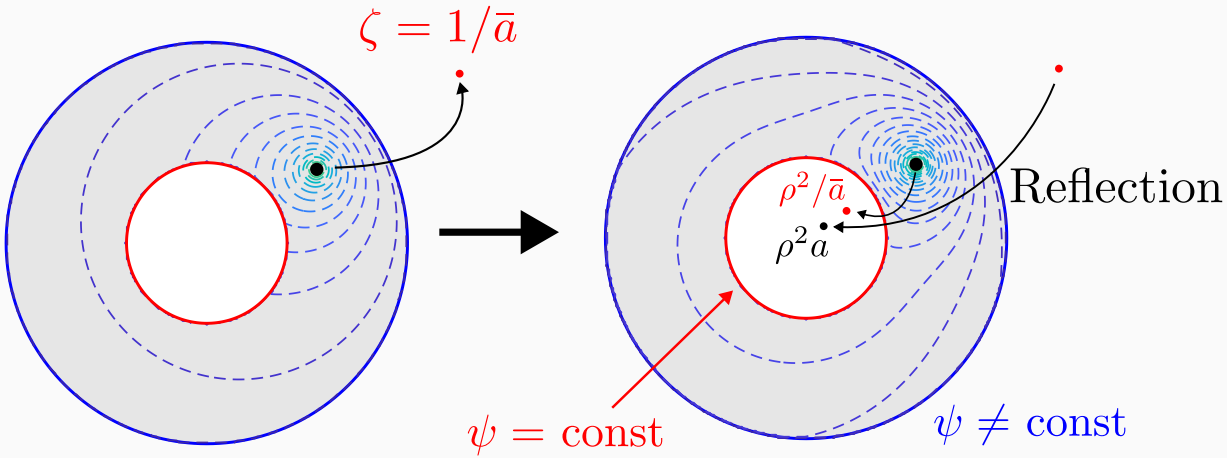


Construction of the complex potential

- Next include the images of $\zeta = a$ and $\zeta = 1/\overline{a}$ in the inner circle.
- Image in the unit circle: $\zeta = 1/\overline{a}$. Image in the inner circle: $\zeta = \rho^2/\overline{a}$.

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}}(\log(\zeta - a) - \log(\zeta - 1/\overline{a}) \\ - \log(\zeta - \rho^2/\overline{a}) + \log(\zeta - \rho^2 a)). \tag{10}$$

- The boundary condition is now satisfied on C_1, but not on C_0.

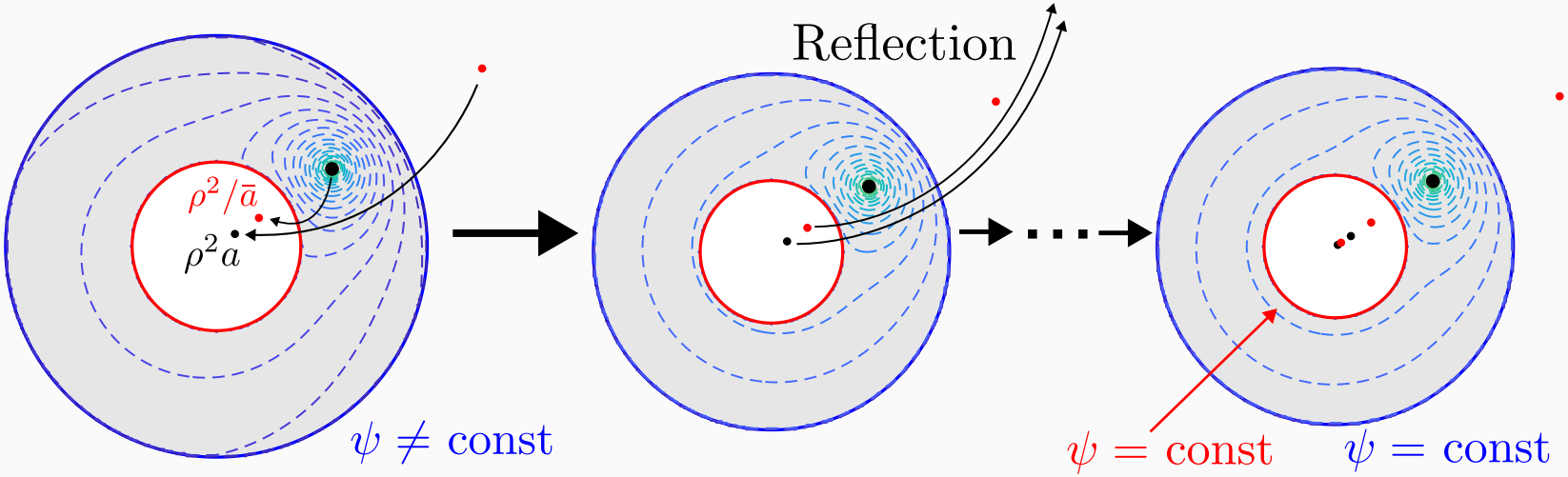


Construction of the complex potential

- Next take images in C_0.

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}}(\log(\zeta - a) - \log(\zeta - 1/\overline{a}) \\ - \log(\zeta - \rho^2/\overline{a}) + \log(\zeta - \rho^2 a)) \\ + \log(\zeta - a/\rho^2) - \log(\zeta - 1/\rho^2\overline{a}) \tag{11}$$

- Repeating this procedure, the two boundary conditions are satisfied more and more accurately.



Construction of the complex potential

- Collecting the $+$ and $-$ terms,

$$
\begin{aligned}
w(\zeta) &= \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{(\zeta-a)(\zeta-\rho^2 a)(\zeta-a/\rho^2)(\zeta-\rho^4 a)}{(\zeta-1/\overline{a})(\zeta-\rho^2/\overline{a})(\zeta-1/\rho^2\overline{a})(\zeta-\rho^4/\overline{a})} \cdots \right) \\
&= \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{(\zeta-a)(\zeta-\rho^2 a)(\zeta-\rho^4 a)}{(\zeta-1/\overline{a})(\zeta-\rho^2/\overline{a})(\zeta-\rho^4/\overline{a})} \cdots \right) \\
&+ \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{(\zeta-a/\rho^2)(\zeta-a/\rho^4)}{(\zeta-1/\rho^2\overline{a})(\zeta-1/\rho^4\overline{a})} \cdots \right) \\
&= \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{(\zeta-a)(1-\rho^2 a/\zeta)(1-\rho^4 a/\zeta)}{(\zeta-1/\overline{a})(1-\rho^2/\overline{a}\zeta)(1-\rho^4/\overline{a}\zeta)} \cdots \right) \\
&+ \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{(1-\rho^2\zeta/a)(1-\rho^4\zeta/a)}{(1-\rho^2\zeta\overline{a})(1-\rho^4\zeta\overline{a})} \cdots \right) + c_0(a,\overline{a})
\end{aligned}
$$

Looking at the numerator and denominator,

$$
\begin{aligned}
&\text{(num.)} \sim -(1-\zeta/a)(1-\rho^2\zeta/a)(1-\rho^2 a/\zeta)(1-\rho^4\zeta/a)(1-\rho^4 a/\zeta)\cdots \\
&\text{(den.)} \sim -(1-\overline{a}\zeta)(1-\rho^2/\zeta\overline{a})(1-\rho^2\zeta\overline{a})(1-\rho^4\zeta\overline{a})(1-\rho^4/\zeta\overline{a})\cdots
\end{aligned}
$$

The Schottky–Klein prime function [2]

In view of the above, we define

$$\omega(\zeta,\alpha) := -\frac{\alpha}{C^2}P(\zeta/\alpha,\rho), \tag{12}$$

$$P(\zeta,\rho) := (1-\zeta)\prod_{n=1}^{\infty}(1-\rho^{2n}\zeta)(1-\rho^{2n}\zeta^{-1}) \tag{13}$$

The complex potential can then be written as

$$w(\zeta) = \frac{\Gamma}{2\pi\mathrm{i}}\log\left(\frac{\omega(\zeta,a)}{|a|\omega(\zeta,1/\overline{a})}\right) = \frac{\Gamma}{2\pi\mathrm{i}}\log\left(\frac{|a|P(\zeta/a,\rho)}{P(\zeta\overline{a},\rho)}\right) \tag{14}$$

- The function $\omega(\zeta,\alpha)$: Schottky–Klein prime function
- It plays a central role in complex analysis and potential theory in multiply connected domains. [2]
- The function $\omega(\zeta,\alpha)$ has the quasi-periodicity familiar from elliptic theta functions.

Properties of $P(\zeta)$

$$P(\zeta,\rho) := (1-\zeta)\prod_{n=1}^{\infty}(1-\rho^{2n}\zeta)(1-\rho^{2n}\zeta^{-1}) \tag{15}$$

Properties

- It has a unique zero in the annulus, at $\zeta = 1$.
- Inversion

$$P(\zeta^{-1},\rho) = -\zeta^{-1}P(\zeta,\rho) \tag{16}$$

- Quasi-periodicity

$$P(\rho^2\zeta,\rho) = -\zeta^{-1}P(\zeta,\rho). \tag{17}$$

A point vortex in the annulus

Complex potential

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{\omega(\zeta, a)}{|a|\omega(\zeta, 1/\overline{a})}\right) = \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{|a|P(\zeta/a, \rho)}{P(\zeta\overline{a}, \rho)}\right) \tag{18}$$

(Why $\mathrm{Im}[w(\zeta)] = \text{const}$ on the boundary)

1. On C_0, $\overline{\zeta} = 1/\zeta$, so

$$\begin{aligned}
\overline{w(\zeta)} &= -\frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{|a|P(\overline{\zeta}/\overline{a}, \rho)}{P(\overline{\zeta}a, \rho)}\right) = -\frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{|a|P(1/\overline{a}\zeta, \rho)}{P(a/\zeta, \rho)}\right) \\
&= -\frac{\Gamma}{2\pi \mathrm{i}} \log\left(|a|\frac{1/\zeta\overline{a} \cdot P(\zeta\overline{a}, \rho)}{a/\zeta \cdot P(\zeta/a, \rho)}\right) = -\frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{P(\zeta\overline{a}, \rho)}{|a|P(\zeta/a, \rho)}\right) \\
&= w(\zeta).
\end{aligned} \tag{19}$$

Hence $\mathrm{Im}[w(\zeta)] = 0$ on $\zeta \in C_0$. The argument on $\zeta \in C_1$ is similar.

Computational method

Computation

- Set the truncation index $N_{\max}$.
- Truncate $P(\zeta, \rho)$ as follows:

$$P(\zeta, \rho) \approx (1 - \zeta) \prod_{n=1}^{N_{\max}} (1 - \rho^{2n}\zeta)(1 - \rho^{2n}\zeta^{-1}) \tag{20}$$

- Define

$$w(z) = \phi + \mathrm{i}\psi = \frac{1}{2\pi\mathrm{i}} \log\left(\frac{\omega(z, a)}{|a|\omega(z, 1/\overline{a})}\right) = \frac{1}{2\pi\mathrm{i}} \log\left(\frac{|a|P(z/a, \rho)}{P(z\overline{a}, \rho)}\right) \tag{21}$$

- The code is in "`03_schottky_klein_annulus.ipynb`"

Streamlines and velocity field

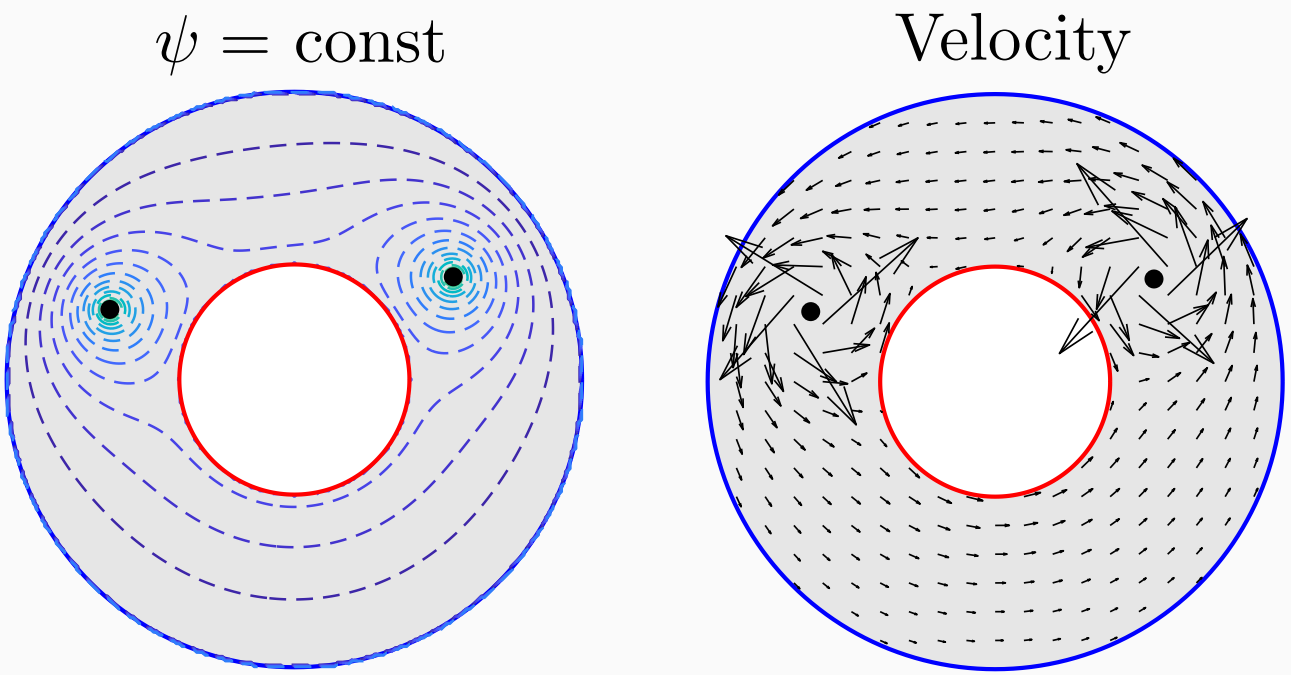


Figure 2: Streamlines and velocity field of two point vortices in an annulus.

Exercise: conformal map

- $$z = g(\zeta) = -\frac{P(\zeta, \rho)}{P(-\zeta, \rho)} \tag{22}$$
- $$z = g(\zeta) = \frac{P(\zeta/a, \rho)P(\zeta\overline{a}, \rho)}{P(\zeta/b, \rho)P(\zeta\overline{b}, \rho)} \tag{23}$$
- $$z = g(\zeta) = \frac{|b|}{|a|}\frac{P(\zeta/a, \rho)P(\zeta\overline{b}, \rho)}{P(\zeta/b, \rho)P(\zeta\overline{a}, \rho)} \tag{24}$$

Mapping

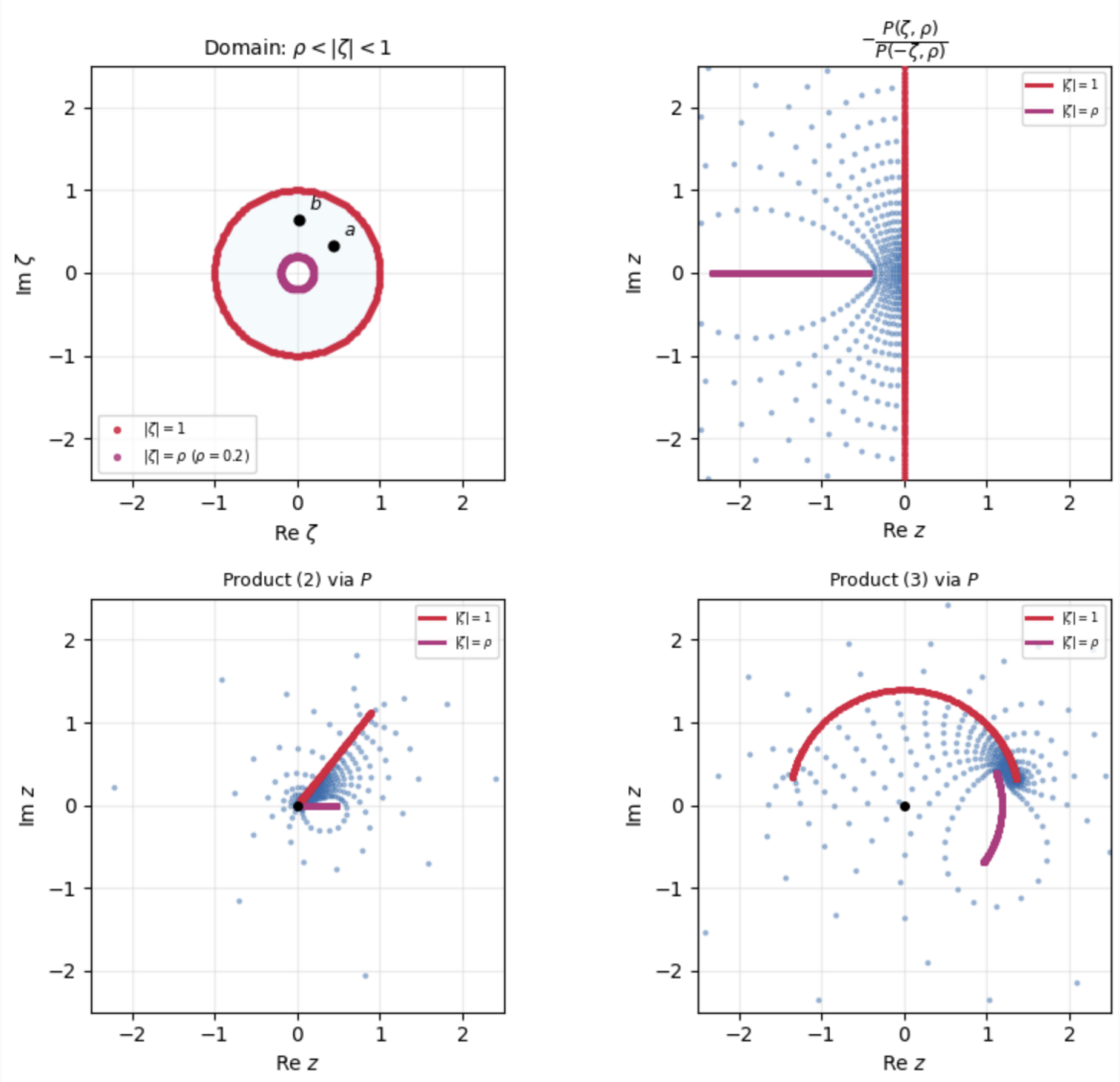


Figure 3: `03_schottky_klein_annulus.ipynb`

PINNs for conformal mapping between simply connected domains

Analytical examples of conformal mapping

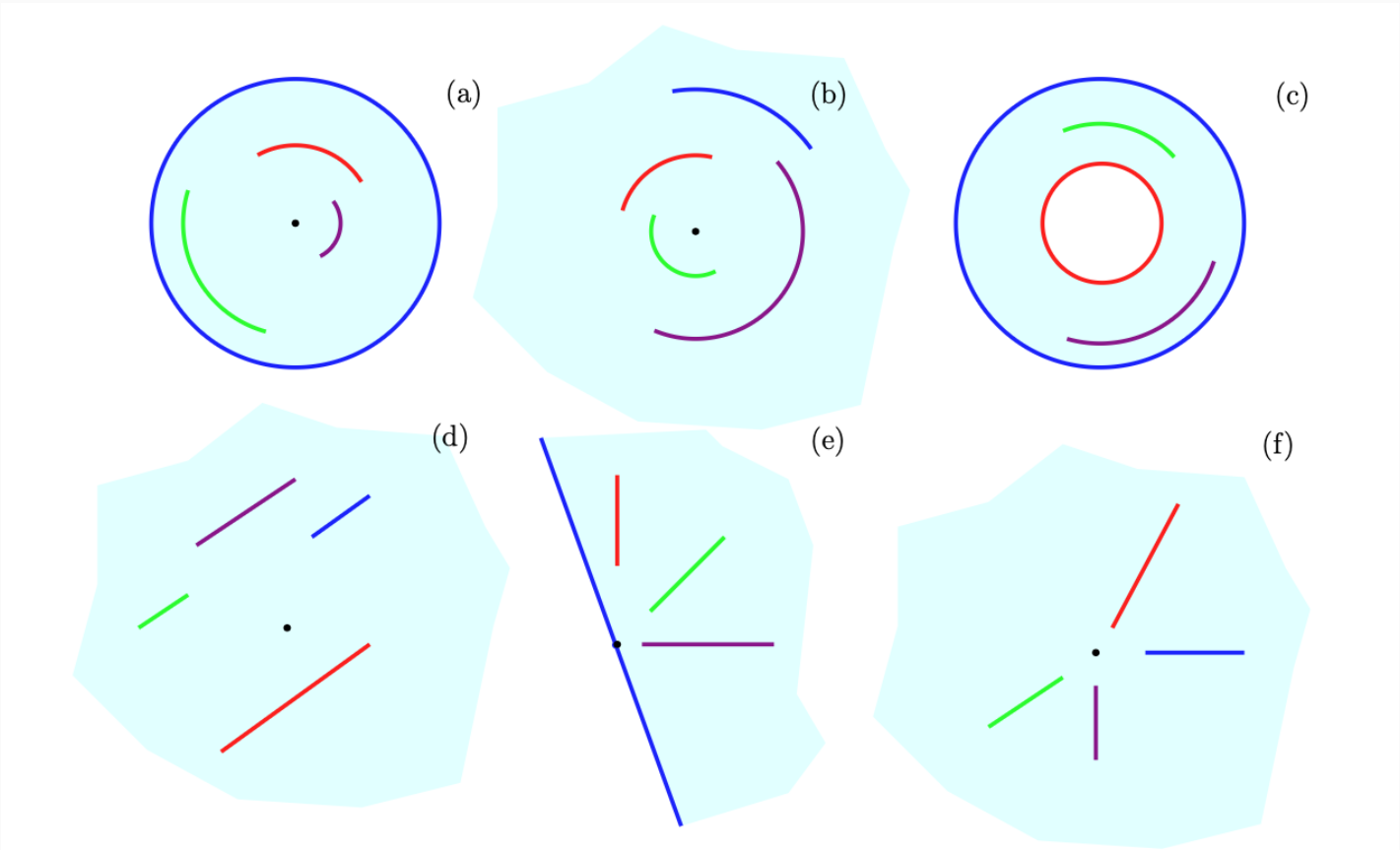


Figure 6.13. *The basic conformal slit mappings from a quadruply connected circular domain such as that in Figure 6.12. (a) A bounded circular slit map (6.94); (b) an unbounded circular slit map (6.95); (c) an annular slit map (6.96); (d) a parallel slit map (6.97); (e) a Cayley-type map (6.98); and (f) a radial slit map (6.99).*

Figure 4: Analytical examples of conformal mapping.

Ways to compute a conformal map

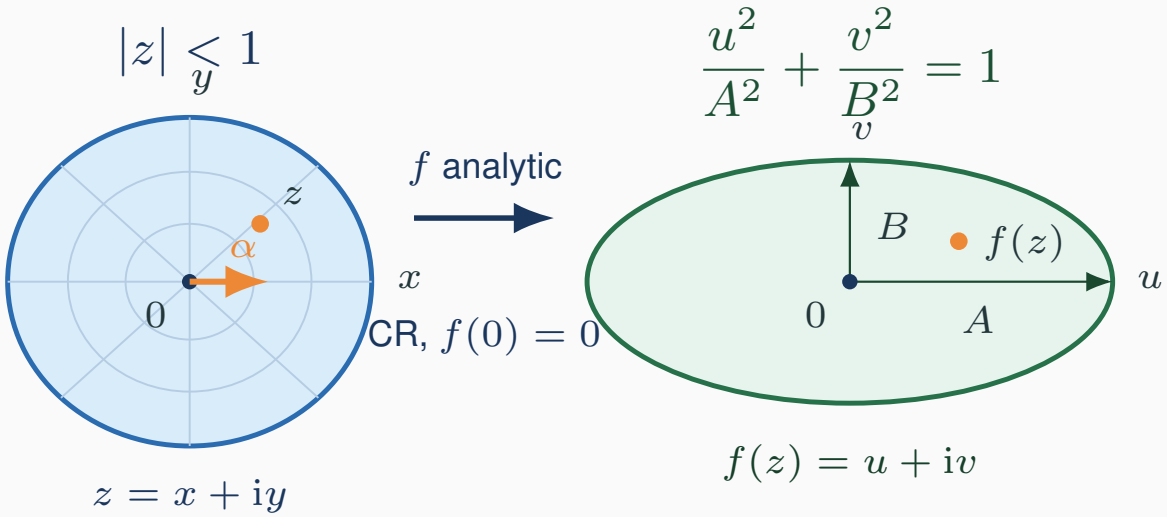
- **Closed form** (today): Möbius maps, powers, $\log$ (the channel).
- **Schwarz–Christoffel**: polygons, via a product of power maps [3] (Driscoll and Trefethen).
- **Polycircular-arc maps**: boundaries made of circular arcs [1] (Bjørstad and Grosse).
- **Integral equations**: Theodorsen, Kerzman–Stein, Wegmann [6] (Wegmann).
- **Zipper**: a geometric composition of slit maps [4] (Marshall and Rohde).
- **Rational approximation** (AAA least squares) from boundary samples [5] (Trefethen).
- **PINNs**: Cauchy–Riemann plus the boundary, as a loss (next frames).

PINNs for conformal mapping

Demo problem (notebook `04_conformal_map_pinn`)

Find $f = u + \mathrm{i}v$ on the unit disk $|z| < 1$ such that

- f is **analytic**: Cauchy–Riemann $u_x = v_y$, $u_y = -v_x$;
- $|z| = 1$ is sent to the ellipse $\dfrac{u^2}{A^2} + \dfrac{v^2}{B^2} = 1$;
- $f(0) = 0$ and $f'(0) = \alpha > 0$ (orientation; α is the conformal radius).



Riemann mapping $\Rightarrow$ such an f exists and is unique for fixed (A, B) and this normalization.

PINN loss = CR + boundary

We do not know the exact solution f. We need to approximate it by a neural network f_θ. Represent f_θ by a network and minimize

$$\mathcal{L} = w_{\mathrm{CR}} \underbrace{\mathbb{E}_{z\in\mathbb{D}}\left[(u_x - v_y)^2 + (u_y + v_x)^2\right]}_{\text{CR / analyticity}} + w_{\mathrm{BC}} \underbrace{\mathbb{E}_{|z|=1}\left[\left(u/A\right)^2 + \left(v/B\right)^2 - 1\right]^2}_{\text{ellipse boundary}}. \tag{25}$$

- Same idea as this morning's Poisson PINN: PDE residual + boundary residual.
- No mesh of the ellipse is required—only samples in the disk and on $|z| = 1$.

Hard constraints

Architecture

Use the power-series style ansatz

$$f(z) = \alpha z + z^2\, N_\theta(z), \qquad \alpha > 0 \text{ trainable} \tag{26}$$

(complex multiplication). Then $f(0) = 0$ and $f'(0) = \alpha$ hold **exactly**. Holomorphicity is still soft: a general real N_θ need not be holomorphic, so the CR loss remains.

Do not force $\alpha = 1$

The conformal radius of a fixed ellipse with semi-axes $A \neq B$ is **not** 1. Fixing $f'(0) = 1$ and the ellipse size at once is generally inconsistent—training stalls or the image leaves the target curve.

Hard constraints

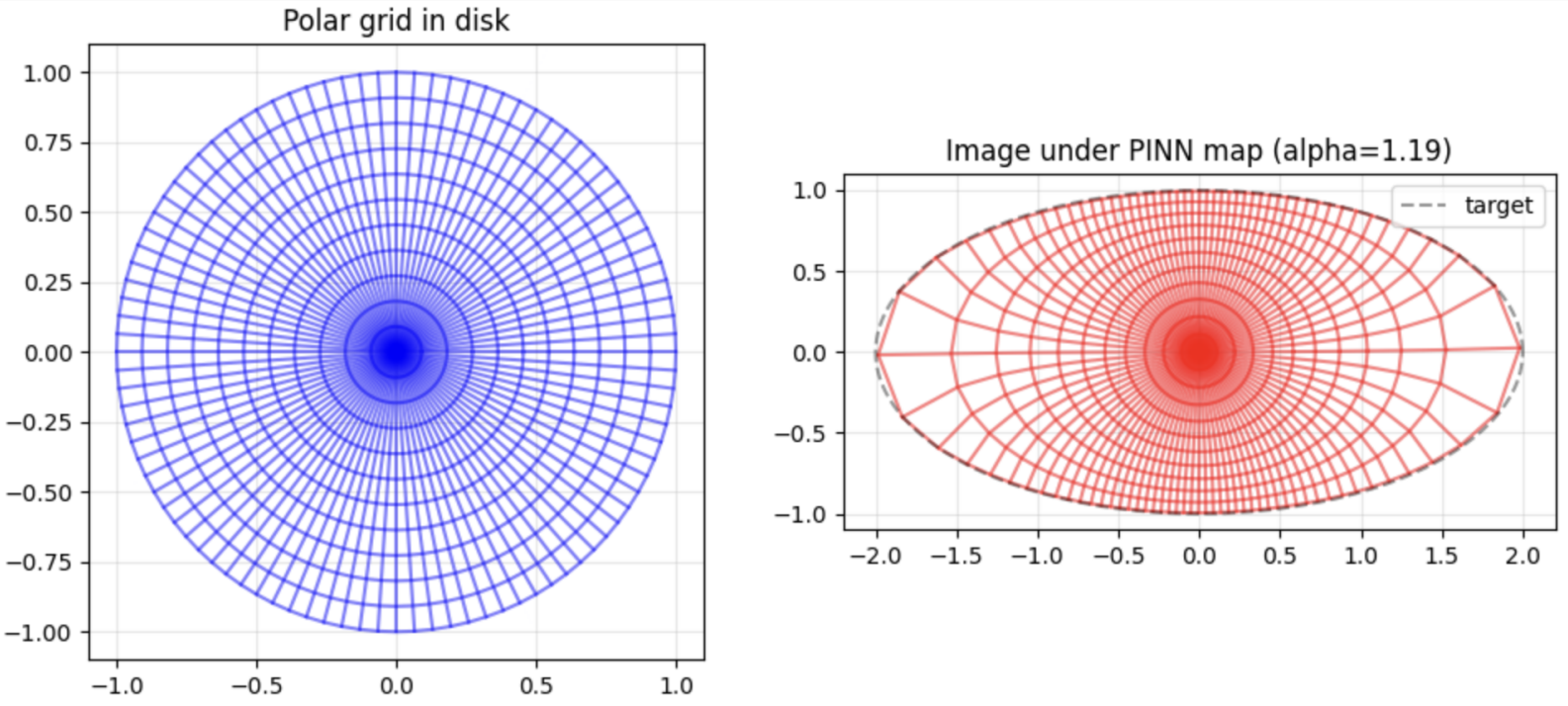


Figure 5: Original domain (left) and the target domain (ellipse)

What the network outputs

The unknown is not P. It is the regular part of the vortex,

$$w(\zeta) = \frac{1}{2\pi \mathrm{i}} \log(\zeta - a) + \hat{w}(\zeta), \tag{27}$$

with $a = 0.5$ and $\rho = 0.3$. $\hat{w} = u + \mathrm{i}v$ is an ordinary MLP, $(x, y) \mapsto (u_\theta, v_\theta)$, with $\tanh$ activations. In the notebook: width 64, depth 5.

The logarithm is written by hand. The network is not asked to fit the singularity at $\zeta = a$. Nothing in the architecture forces $u + \mathrm{i}v$ to be holomorphic. The code is `03_vortex_hat_w.ipynb`.

Cauchy–Riemann as a loss

Holomorphicity of $\hat{w}$ is the pair

$$\partial_x u = \partial_y v, \qquad \partial_y u = -\partial_x v. \tag{28}$$

Both derivatives come from `autograd`, as on Day 1. At collocation points in the annulus,

$$\mathcal{L}_{\mathrm{CR}} = \mathrm{mean}\big((\partial_x u - \partial_y v)^2 + (\partial_y u + \partial_x v)^2\big). \tag{29}$$

What this term can do

It penalises a non-holomorphic $\hat{w}$. It does not make the boundary circles into streamlines.

Constant imaginary part on each circle

$$\operatorname{Im} w = -\frac{1}{2\pi} \log |\zeta - a| + v_\theta(\zeta). \tag{30}$$

On $|\zeta| = 1$ and on $|\zeta| = \rho$, the loss is the variance of $\operatorname{Im} w$. The two constants are not required to coincide.

Gauge

The mean of $\operatorname{Im} w$ on the outer circle is set to 0, and the mean of u_θ there is set to 0. The constant on the inner circle is predicted, not prescribed.

The product does not enter this loss.

Circulation on the inner circle

$\hat{w}$ is single-valued, so it contains no term $c \log \zeta$. The loss still records the period

$$\oint_{|\zeta|=\rho} \mathrm{d}u_\theta, \tag{31}$$

which stays at the size of the quadrature.

The explicit logarithm has circulation 0 on $|\zeta| = \rho$ and circulation 1 on $|\zeta| = 1$, because a lies between the two circles. A holomorphic $\hat{w}$ does not change those two numbers.

Check

What is being compared

1. Truncate $P(\zeta, \rho)$ and form w. This is the reference, after training.
2. Build w from $\log(\zeta - a)$ and the trained $\hat{w}$.
3. Compare $\operatorname{Im} w$, and the circulation on each circle.

One run, $\rho = 0.3$, $a = 0.5$

After 3000 steps the relative L^2 error of $\operatorname{Im} w$ is about 0.03. The inner constant is about 0.111, against 0.110 from the product.

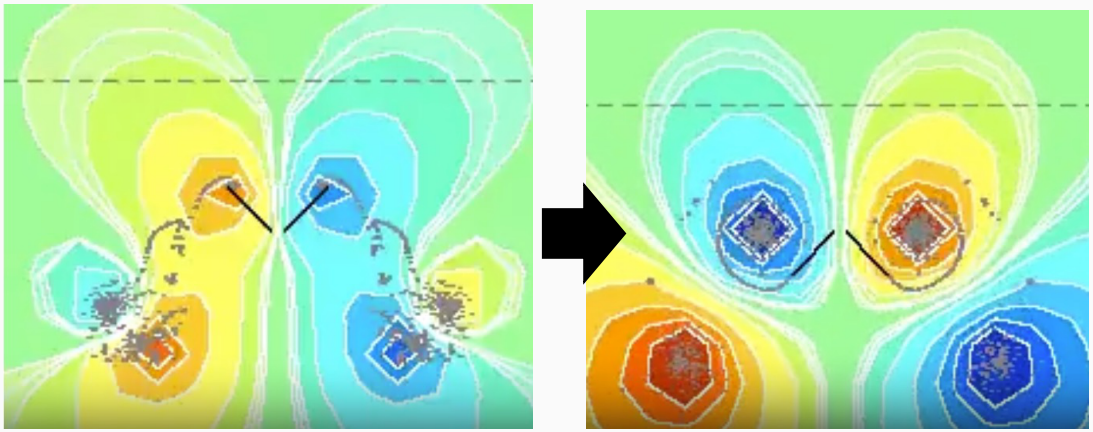
Conclusions

- One hole is an annulus. The reflections are the product $P(\zeta, \rho)$, cut at $N_{\max}$.
- The unit test is the inversion identity at sample points.
- The point vortex is the test case: $\operatorname{Im} w$ is constant on each circle.
- The network learns $\hat{w}$, not P. The product is the check after training.
- Circulation around the inner circle is 0. Around the outer circle it is 1.

Summary

- We introduced a function $P(\zeta, \rho)$ to solve a problem in doubly connected domains.
- We have solved a potential problem, but it can be used for solving many problems in such domains.

Insect hovering



https://fluid.hiroshima-u.ac.jp/%E7%A0%94%E7%A9%B6/

Figure 6: Streamlines and velocity field of two point vortices in an annulus.

References

[1] Bjørstad, P., Grosse, E., 1987. Conformal mapping of circular arc polygons. SIAM Journal on Scientific and Statistical Computing 8, 19–32.

[2] Crowdy, D., 2020. Solving Problems in Multiply Connected Domains. CBMS-NSF regional conference series in applied math, Society for Industrial and Applied Mathematics. URL: https://books.google.co.jp/books?id=A5hSzQEACAAJ.

[3] Driscoll, T.A., Trefethen, L.N., 2002. Schwarz–Christoffel Mapping. Cambridge University Press.

[4] Marshall, D.E., Rohde, S., 2007. Convergence of a variant of the zipper algorithm for conformal mapping. SIAM Journal on Numerical Analysis 45, 2577–2609.

[5] Trefethen, L.N., 2025. Numerical conformal mapping. arXiv preprint arXiv:2507.14872 .

[6] Wegmann, R., 2005. Methods for numerical conformal mapping, in: Kühnau, R. (Ed.), Handbook of Complex Analysis: Geometric Function Theory, Vol. 2. Elsevier, pp. 351–477.

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